

Abu Jafar Saifullah

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RESEARCH INTERESTS

AI for Software Engineering, Human-Centered Software Engineering, Web Accessibility

EDUCATION

University of Dhaka

M.Sc. in Software Engineering; GPA: 3.92/4.00

B.Sc. in Software Engineering; GPA: 3.89/4.00

Dhaka, Bangladesh

Apr 2025 – Sep 2026

Jan 2019 – Dec 2023

Notre Dame College

Higher Secondary Certificate; GPA: 5.00/5.00

Dhaka, Bangladesh

Jun 2016 – Jun 2018

PUBLICATIONS

Abu Jafar Saifullah, Tasmia Zerin, Zerina Begum, Kazi Sakib. “NavA11y: A Dynamic Analysis Approach for WCAG 2.4 Focus-Behavior Evaluation.” In *Proceedings of the 21st International Conference on Evaluation of Novel Approaches to Software Engineering (ENASE 2026)*, vol. 2, pp. 1019–1030. SCITEPRESS. Benidorm, Spain. *Full paper; presented in person, May 2026*. DOI: [10.5220/0015045700004015](https://doi.org/10.5220/0015045700004015) [\[Paper\]](#) [\[Code\]](#)

Method: Drives a real browser through each page’s keyboard tab order and, at every focus stop, checks that focus lands where expected, is visibly indicated, and is not hidden behind other content.

Results: 100% precision and recall on a 22-page labeled dataset; 2,947 violations on 26 production websites at 90% precision, validated by two independent expert raters; first tool to cover all five WCAG 2.4 focus-behavior criteria.

RESEARCH EXPERIENCE

DSSE Research Group, IIT, University of Dhaka

Dhaka, Bangladesh

Graduate Researcher; supervisor Dr. Zerina Begum, co-supervisor Dr. Kazi Sakib

Apr 2025 – Present

- M.Sc. thesis (submitted): *Detecting and Repairing Keyboard Focus-Behavior Violations through Dynamic Analysis*. Built **NavA11y** for detection and **RepairA11y** for repair.
- Designed NavA11y’s two-phase analysis: page level drives real keyboard navigation and compares observed vs. expected focus order; element level checks each focused element is visible, has a focus indicator, and is not hidden.
- Extended the GDS Accessibility Tool Audit’s 6 focus test pages with 16 new pages into a 22-page labeled dataset (*Dd*) covering all five WCAG 2.4 focus-behavior SCs, including pass, boundary, and obscurity cases; released for evaluating future tools.

SEMERU Lab, William & Mary

Remote

Research Intern, supervised by Dr. Denys Poshyvanyk

Aug 2023 – Dec 2023

- Built a programming language-specific tokenizer to improve NLP model predictions for software engineering tasks, contributing to the lab’s research on intelligent software tools.

INDUSTRY EXPERIENCE

Optimizely, Content Marketing Platform

Dhaka, Bangladesh

Software Engineer II (promoted from SE I, Apr 2026)

Apr 2024 – Present, Full-time

- Delivered end-to-end integration of an internal AI SDK by moving its service from Express to Hono, adding the AWS CodeArtifact publishing pipeline, and instrumenting AI-context events across **10+ product surfaces**.
- Shipped the platform’s block-based AI document UI (Adaptive-Cards-style): backend block-context extraction for LLM prompts, frontend block-form interaction, and AI resource rendering.
- Platform engineering: led migration to **Axiom (@optiaxiom/react)** with **26 merged upstream PRs** and the adopted Migration/Icon Contribution Guides; rolled out Dark Mode across **15+ surfaces**; co-built **Spotlight** global search (winner, **Optimizely Global Hackathon 2025**); reviewed **300+ pull requests**.

Orion Informatics Ltd.

Dhaka, Bangladesh

Software Engineer

Jul 2023 – Mar 2024, Full-time

- Built data visualisation and reporting features for an Attendance Management System; developed a two-service ASP.NET Core webhook relay for TradeStation alerts deployed on Azure.

RESEARCH & ACADEMIC PROJECTS

NavA11y | *Node.js, Playwright, Chromium, GitHub Actions* | [Code](#) | *ENASE 2026*

- Dynamic WCAG 2.4 accessibility testing tool, the first to cover all five focus-behavior success criteria. Drives a real browser through each page's keyboard navigation and inspects what is actually rendered at every focus stop. 100% precision/recall on a 22-page labeled benchmark; 2,947 violations on 26 production sites; 90% precision confirmed by independent expert raters; dataset released for community benchmarking.

RepairA11y | *Node.js, Playwright, Vitest* | *M.Sc. Thesis, 2026*

- Repairs WCAG 2.4 focus-behavior violations from the runtime evidence NavA11y records: writes a typed patch (a CSS rule or an attribute change) with deterministic rules or a language model, applies it in a live browser, and re-runs NavA11y's checks. On 32 fixture cases, rules fixed 29 and the model 25, with no new focus-behavior violation; runtime evidence raised the model from 5 to 7 of 7 realistic focus-appearance cases.

Clippy | *Angular, Node.js, AWS S3* | [GitHub](#) | *ICSE SCORE 2023, Top 6*

- Smart PDF reader for researchers with knowledge-graph extraction, citation-network visualization, and paper recommendation; Top 6 finalist in the ICSE 2023 Student Contest on Software Engineering (SCORE).

BPE Tokenizer | *Python, Flask, HuggingFace* | [PyPI](#)

- Byte Pair Encoding tokenizer implemented from scratch and published as `python-bpe-tokenizer` on PyPI; includes a Flask web interface for training on custom corpora. Built as direct precursor to SEMERU Lab tokenizer research.

ML Bug Mining (SZZ) | *Python, GitPython, pandas, scikit-learn* | *M.Sc. Coursework*

- Mined ML-related bugs across 9 open-source repositories using the SZZ algorithm; identified bug-fixing and bug-introducing commits via keyword heuristics and git blame, filtered for ML-library-touching code.

OTHER PROJECTS

Blueberry Auto | *React, Hono, Supabase, Vercel* | [Live](#) | *Lead Developer, 2025 – Present*

- Vehicle auction and marketplace platform built as a pnpm monorepo (frontend, Hono API, Supabase backend, crawler); leading end-to-end build as the contracted developer (**150+ merged PRs**).

TECHNICAL SKILLS

Research Methods: Dynamic Program Analysis, Mining Software Repositories (SZZ), Controlled User Studies, Statistical Evaluation (Kendall τ , Cohen's κ , PABAK), Inter-rater Agreement

Languages: TypeScript, JavaScript, Python, Java, C#, C++

Web & Frontend: React 18/19, Next.js, Angular, Node.js, Hono, Express, FastAPI, Spring Boot 3, ASP.NET Core

Testing & Automation: Playwright, axe-core, Vitest, Karma, Mocha, Chai, Sinon, Selenium

ML / NLP: TensorFlow, HuggingFace Transformers, BPE Tokenization (own impl.), TextRank, TF-IDF, scikit-learn, pandas

DevOps / Cloud: AWS CodeArtifact, AWS S3, Azure, Cloudflare Workers, Docker, NGINX, GitHub Actions

Databases: MongoDB, PostgreSQL (Supabase), SQL Server, Firebase, Redis

AWARDS & ACHIEVEMENTS

ICT Division Fellowship, Government of Bangladesh: research funding for M.Sc. thesis (Grant No. 56.00.0000.052.33.002.24-61, Apr 2026)

Winner, Optimizely Global Hackathon 2025 (company-wide international hackathon; team built Spotlight, shipped to CMP)

Top 6, ICSE SCORE 2023 (international undergraduate SE competition)